EXERCISE – 5.3

INFERENCE ABOUT DIFFERENCE OF MEANS WHEN SIGMAS ARE KNOWN

1. A credit card watchdog group claims that there is a difference in the mean credit card debts of people in California and Florida. The sample means of a random survey of 250 people from each state are $\overline{x_1} = \$3060$ *and* $\overline{x_2} = \$2960$. The two samples are independent. Assume that $\sigma_1 = \$960$ for California and $\sigma_2 = \$845$ for Florida. Do the results support the group's claim? Use $\alpha = 0.05$. Use all the three approaches

$(a)\,Critical-Value\ Approach\ (b)\ P-value\ Approach\ \ (c)\,Confidence\ Interval\ Approach$

Ans $(H_0: \mu_1 = \mu_2,\ H_1: \mu_1 \neq \mu_2,\ z = 1.85,\ P-value = 0.0644,\ CI: (-58.52 \leq \mu_1 - \mu \leq 258.52),\ Conclusion: do\ not\ reject\ H_0)$

2. You want to buy a car, and a salesperson tells you that the mean running costs for Model A and Model B are equal. You research the running costs. The mean running cost of 24 Model A cars is \$5 per kilometer. Assume the population standard deviation is \$1.50. The mean running cost of 26 Model B cars is \$6.5 per kilometer. Assume the population standard deviation is \$2.50. At $\alpha = 0.05$, can you reject the salesperson's claim? Use all the three approaches

$(a)\,Critical-Value\ Approach\ (b)\ P-value\ Approach\ \ (c)\,Confidence\ Interval\ Approach$

Ans $(H_0: \mu_1 = \mu_2,\ H_1: \mu_1 \neq \mu_2,\ z = -2.60,\ P-value = 0.0093, CI: (-2.63 \leq \mu_1 - \mu \leq -0.37),\ Conclusion: reject\ H_0)$

3. A non-governmental organization wants to choose between two regions in a state to initiate a campaign for rainwater harvesting. A researcher claims that Region A receives lesser rainfall than Region B. To test the regions, the average rainfall is calculated for 60 days in each region. The mean rainfall in Region A is 700 millimeters. Assume the population standard deviation is 60 millimeters. The mean rainfall in Region B is 725 millimeters. Assume the population standard deviation is 66 millimeters. At $\alpha = 0.01$, can the organization support the researcher's claim? Use all the three approaches

$(a)\,Critical-Value\ Approach\ (b)\ P-value\ Approach\ \ (c)\,Confidence\ Interval\ Approach$

Ans $(H_0: \mu_1 \geq \mu_2,\ H_1: \mu_1 < \mu_2,\ z = -2.17,\ P-value = 0.015,\ CI: (\mu_1 - \mu \leq 1.82),\ Conclusion: do\ not\ reject\ H_0)$

4. To compare customer satisfaction with holiday gift purchases of digital gear from online and walk-in retailers, a researcher randomly selects 30 customer ratings of online retailers and 31 customer ratings of walk-in retailers. The mean customer rating of online retailers is 90 out of 100. Assume the population standard deviation is 3.4. The mean customer rating of walk-in retailers is 88 out of 100. Assume the population standard deviation is 3.5. At $\alpha = 0.01$, can the researcher support the claim that the mean customer rating of online retailers is greater than the mean customer rating of walk-in retailers? Use all the three approaches

$(a)\,Critical-Value\ Approach\ (b)\ P-value\ Approach\ \ (c)\,Confidence\ Interval\ Approach$

Ans $(H_0: \mu_1 \leq \mu_2,\ H_1: \mu_1 > \mu_2,\ z = 2.26,\ P-value = 0.012,\ CI: (\mu_1 - \mu \geq -0.058),\ Conclusion: do\ not\ reject\ H_0)$

5. Gibbs Baby Food Company wishes to compare the weight gain of infants using its brand versus its competitor's. A sample of 40 babies using the Gibbs products revealed a mean weight gain of 7.6 pounds in the first three months after birth. For the Gibbs brand, the population standard deviation of the sample is 2.3 pounds. A sample of 55 babies using the competitor's brand revealed a mean increase in weight of 8.1 pounds. The population standard deviation is 2.9 pounds. At the $\alpha = 0.05$ significance level, can we conclude that babies using the Gibbs brand gained less weight? Test the claim using

$(a)\,Critical-Value\ Approach\ (b)\ P-value\ Approach\ \ (c)\,Confidence\ Interval\ Approach$

Ans $(H_0: \mu_1 \geq \mu_2,\ H_1: \mu_1 < \mu_2,\ z = -0.94,\ P-value = 0.1736,\ CI: (\mu_1 - \mu \leq 0.3811),\ Conclusion: do\ not\ reject\ H_0)$

HAZBER SAMSON

EXERCISE – 5.4

INFERENCE ABOUT DIFFERENCE OF MEANS WHEN SIGMAS ARE UNKNOWN

1. A random sample of 15 observations from the first population revealed a sample mean of 350 and a sample standard deviation of 12. A random sample of 17 observations from the second population revealed a sample mean of 342 and a sample standard deviation of 15. Assuming equal variances, At the 0.10 significance level, is there a difference in the population means? Test the claim using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

Ans $(H_0: \mu_1 = \mu_2,\ H_1: \mu_1 \neq \mu_2,\ z = 1.85,\ P-value = 0.0644,\ CI: (-58.52 \le \mu_1 - \mu \le 258.52),\ Conclusion: do\ not\ reject\ H_0)$

Ans $(t = 1.65, P-value = 0.1093, CI: -0.7548 \le \mu_1 - \mu_2 \le 16.7548,\ Conclusion: do\ not\ reject\ H_0)$

2. For a marketing class term project, Bob is investigating whether college seniors eat less frequently in fast-food chains than college freshmen. He asked 11 freshmen and 11 seniors to keep track of how many times they ate in a fast-food restaurant during the month of October. Assuming equal variances, can you conclude that the mean is significantly smaller for college seniors at the 5 percent level of significance? Test the claim using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

Ans $(t = 1.55,\ P-value =$ $,\ do\ not\ reject\ H_0,\ CI: (-0.7548, 16.7548))$

3. A recent study compared the time spent together by single- and dual-earner couples. According to the records kept by the wives during the study, the mean amount of time spent together watching television among the single-earner couples was 61 minutes per day, with a standard deviation of 15.5 minutes. For the dual-earner couples, the mean number of minutes spent watching television was 48.4 minutes, with a standard deviation of 18.1 minutes. At the .01 significance level, can we conclude that the single-earner couples on average spend more time watching television together? There were 15 single-earner and 12 dual-earner couples studied. Test the claim using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

Ans $(t = 1.55,\ P-value =$ $,\ do\ not\ reject\ H_0,\ CI: (-0.7548, 16.7548))$

4. For a marketing class term project, Bob is investigating whether college seniors eat less frequently in fast-food chains than college freshmen. He asked 11 freshmen and 11 seniors to keep track of how many times they ate in a fast-food restaurant during the month of October. Assuming equal variances, can you conclude that the mean is significantly smaller for college seniors at the 5 percent level of significance? *Test the claim using*

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

Seniors	10	5	15	13	5	7	18	8	19	9	8
Freshmen	16	9	17	14	15	11	18	12	7	16	20

Ans $(t = 1.55,\ P-value =$ $,\ do\ not\ reject\ H_0,\ CI: (-0.7548, 16.7548))$

5. Ms. Lisa is the budget director for Nexus Media Inc. She would like to compare the daily travel expenses for the sales staff and the audit staff. She collected the following sample information.

Sales ($)	131	135	146	165	136	142	
Audit ($)	130	102	129	143	149	120	139

At the 0.10 significance level, can she conclude that the mean daily expenses are greater for the sales staff than the audit staff? Check the claim using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

Ans $(t = 1.536,\ P-value =$ $,\ \text{Re}\ ject\ H_0,\ CI: (-2.048, 26.468))$

HAZBER SAMSON